Tutorial - 6 (20/2/2026)

propaty Features and prices.

Bedrooms (x_1)	Age (years) (x_2)	price (5100000s) (y)
3	5	2
3	3	3

m_1 m_2

$$CF = J = \|y - xw\|^2 + \lambda \|w\|^2$$

linear equations: $2 = c + 3m_1 + 5m_2$

$3 = c + m_1 3 + 3m_2$

$$\underset{y}{\begin{bmatrix} 2 \\ 3 \end{bmatrix}} = \underset{x}{\begin{bmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{bmatrix}} \underset{w}{\begin{bmatrix} c \\ m_1 \\ m_2 \end{bmatrix}}$$

$$W = (X^T X + \lambda I)^{-1} (X^T y)$$

$$X^T X = \begin{bmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{bmatrix} = \begin{bmatrix} 1+1 & 3+3 & 5+3 \\ 3+3 & 9+9 & 15+9 \\ 5+3 & 15+9 & 25+9 \end{bmatrix} = \begin{bmatrix} 2 & 6 & 8 \\ 6 & 18 & 24 \\ 8 & 24 & 34 \end{bmatrix}$$

$3 \times 2 \quad 2 \times 3$

$$\lambda I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad X^T X + \lambda I = \begin{bmatrix} 2 & 6 & 8 \\ 6 & 18 & 24 \\ 8 & 24 & 34 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2+3 \\ 6+9 \\ 10+9 \end{bmatrix} = \begin{bmatrix} 5 \\ 15 \\ 19 \end{bmatrix}$$

$3 \times 2 \quad 2 \times 1$

$$\begin{bmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{bmatrix}^{-1} = \frac{1}{95} \begin{bmatrix} 89 & -18 & -8 \\ -18 & 41 & -24 \\ -8 & -24 & 21 \end{bmatrix} = \begin{bmatrix} 0.93 & -0.18 & -0.08 \\ -0.18 & 0.43 & -0.25 \\ -0.08 & -0.25 & 0.22 \end{bmatrix}$$

det = 95

$$\Rightarrow W = \begin{bmatrix} 0.93 & -0.18 & -0.08 \\ -0.18 & 0.43 & -0.25 \\ -0.08 & -0.25 & 0.22 \end{bmatrix} \begin{bmatrix} 5 \\ 15 \\ 19 \end{bmatrix} = \begin{bmatrix} 0.43 \\ 0.80 \\ 0.03 \end{bmatrix}$$

$$= \cancel{4.65 - 2.7 - 1.52}$$

$$y^T = 0.43 + 0.80 + 0.03 = 1.26 \times$$

$$\hat{y} = 0.43 + 0.80(3) + 0.03(5)$$

$$= 0.43 + 2.4 + 0.15 = 2.98$$

$$\hat{y} = 0.43 + 0.80(3) + 0.03(3)$$

$$= 0.43 + 2.4 + 0.09 = 2.92$$

$$XW = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{bmatrix} \begin{bmatrix} 0.43 \\ 0.80 \\ 0.03 \end{bmatrix} = \begin{bmatrix} 0.43 + 2.4 + 0.15 \\ 0.43 + 2.4 + 0.09 \end{bmatrix} = \begin{bmatrix} 2.98 \\ 2.92 \end{bmatrix}$$

$2 \times 3 \quad 3 \times 1$

$$y - XW = \begin{bmatrix} 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 2.98 \\ 2.92 \end{bmatrix} = \begin{bmatrix} -0.98 \\ 0.08 \end{bmatrix}$$

$= [0.9604 + 0.0064] + 1[0.1849 + 0.64 + 0.0009]$

$= [0.9668] + [0.8258]$

$= 1.7926$

20/07/25